

Cascade R-CNN: Delving into High Quality Object Detection

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Abstract

In object detection, an intersection over union (IoU) threshold is required to define positives and negatives. An object detector, trained with low IoU threshold, e.g. 0.5, usually produces noisy detections. However, detection performance tends to degrade with increasing the IoU thresholds. Two main factors are responsible for this: 1) overfitting during training, due to exponentially vanishing positive samples, and 2) inference-time mismatch between the IoUs for which the detector is optimal and those of the input hypotheses. A multi-stage object detection architecture, the Cascade R-CNN, is proposed to address these problems. It consists of a sequence of detectors trained with increasing IoU thresholds, to be sequentially more selective against close false positives. The detectors are trained stage by stage, leveraging the observation that the output of a detector is a good distribution for training the next higher quality detector. The resampling of progressively improved hypotheses guarantees that all detectors have a positive set of examples of equivalent size, reducing the overfitting problem. The same cascade procedure is applied at inference, enabling a closer match between the hypotheses and the detector quality of each stage. A simple implementation of the Cascade R-CNN is shown to surpass all single-model object detectors on the challenging COCO dataset. Experiments also show that the Cascade R-CNN is widely applicable across detector architectures, achieving consistent gains independently of the baseline detector strength. The code will be made available at <https://github.com/zhaoweicai/cascade-rcnn>.

1. Introduction

Object detection is a complex problem, requiring the solution of two main tasks. First, the detector must solve the *recognition* problem, to distinguish foreground objects from background and assign them the proper object class labels. Second, the detector must solve the *localization* problem, to assign accurate bounding boxes to different objects. Both of these are particularly difficult because the detector faces

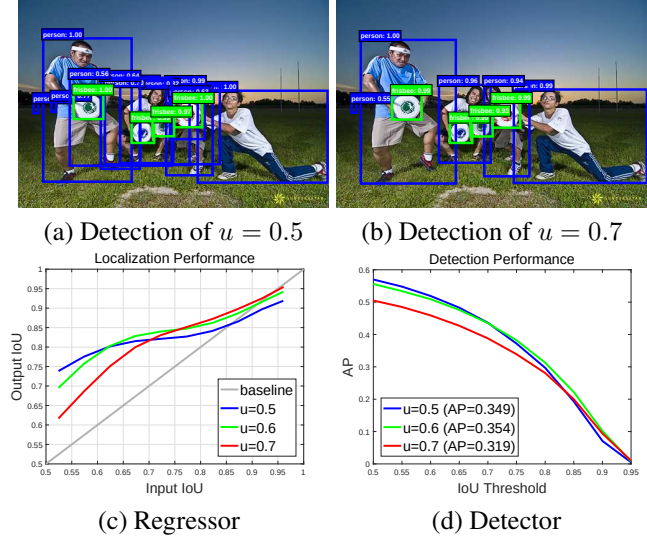


Figure 1. The detection outputs, localization and detection performance of object detectors of increasing IoU threshold u .

many “close” false positives, corresponding to “close but not correct” bounding boxes. The detector must find the true positives while suppressing these close false positives.

Many of the recently proposed object detectors are based on the two-stage R-CNN framework [12, 11, 27, 21], where detection is framed as a multi-task learning problem that combines classification and bounding box regression. Unlike object recognition, an intersection over union (IoU) threshold is required to define positives/negatives. However, the commonly used threshold values u , typically $u = 0.5$, establish quite a loose requirement for positives. The resulting detectors frequently produce noisy bounding boxes, as shown in Figure 1 (a). Hypotheses that most humans would consider close false positives frequently pass the $IoU \geq 0.5$ test. While the examples assembled under the $u = 0.5$ criterion are rich and diversified, they make it difficult to train detectors that can effectively reject close false positives.

In this work, we define the *quality* of an hypothesis as its IoU with the ground truth, and the *quality of the detector* as the IoU threshold u used to train it. The goal is to investi-

级联R-CNN：深入探索高质量目标检测

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摘要

In object detection, an intersection over union (IoU) threshold is required to define positives and negatives. An object detector, trained with low IoU threshold, e.g. 0.5, usually produces noisy detections. However, detection performance tends to degrade with increasing the IoU thresholds. Two main factors are responsible for this: 1) overfitting during training, due to exponentially vanishing positive samples, and 2) inference-time mismatch between the IoUs for which the detector is optimal and those of the input hypotheses. A multi-stage object detection architecture, the Cascade R-CNN, is proposed to address these problems. It consists of a sequence of detectors trained with increasing IoU thresholds, to be sequentially more selective against close false positives. The detectors are trained stage by stage, leveraging the observation that the output of a detector is a good distribution for training the next higher quality detector. The resampling of progressively improved hypotheses guarantees that all detectors have a positive set of examples of equivalent size, reducing the overfitting problem. The same cascade procedure is applied at inference, enabling a closer match between the hypotheses and the detector quality of each stage. A simple implementation of the Cascade R-CNN is shown to surpass all single-model object detectors on the challenging COCO dataset. Experiments also show that the Cascade R-CNN is widely applicable across detector architectures, achieving consistent gains independently of the baseline detector strength. The code will be made available at <https://github.com/zhao-weicai/cascade-rcnn>.

1. 引言

目标检测是一个复杂的问题，需要解决两个主要任务。首先，检测器必须解决recognition问题，以区分前景物体与背景，并为它们分配正确的物体类别标签。其次，检测器必须解决localization问题，为不同物体分配精确的边界框。这两项任务都特别困难，因为检测器面临

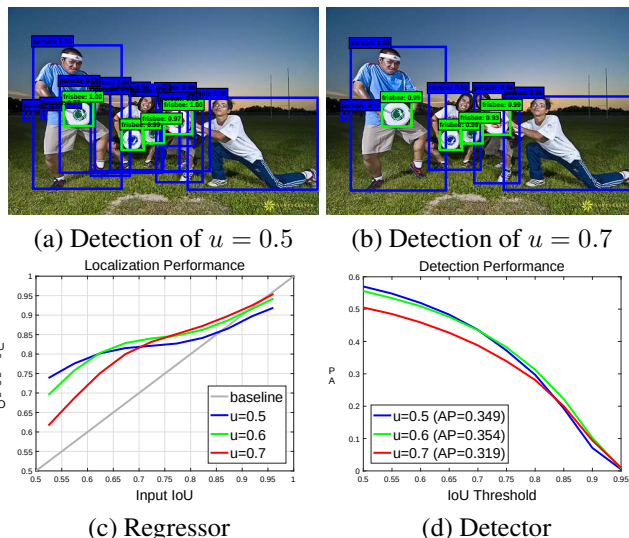


图1. 检测输出、定位以及IoU阈值 u 递增时目标检测器的检测性能。

许多“接近”的假阳性，对应着“接近但不正确”的边界框。检测器必须在抑制这些接近假阳性的同时找出真阳性。

许多最近提出的目标检测器基于两阶段R-CNN框架[12, 11, 27, 21]，其中检测被构建为结合分类与边界框回归的多任务学习问题。与目标识别不同，定义正负样本需要交并比 (IoU) 阈值。然而，常用的阈值 u (通常为 $u = 0.5$) 为正样本设定了相当宽松的要求。如图1(a)所示，由此产生的检测器常生成带有噪声的边界框。大多数人类会判定为近似假阳性的假设经常能通过 $IoU \geq 0.5$ 的测试。虽然基于 $u = 0.5$ 标准收集的样本丰富多样，但这使得训练能有效排除近似假阳性的检测器变得困难。

在这项工作中，我们将一个假设的 $quality$ 定义为其与真实标注的交并比，并将 $quality$ of the detector定义为用于训练该假设的交并比阈值 u 。目标是研究——

gate the, so far, poorly researched problem of learning high quality object detectors, whose outputs contain few close false positives, as shown in Figure 1 (b). The basic idea is that a single detector can only be optimal for a single quality level. This is known in the cost-sensitive learning literature [7, 24], where the optimization of different points of the receiver operating characteristic (ROC) requires different loss functions. The main difference is that we consider the optimization for a given IoU threshold, rather than false positive rate.

The idea is illustrated by Figure 1 (c) and (d), which present the localization and detection performance, respectively, of three detectors trained with IoU thresholds of $u = 0.5, 0.6, 0.7$. The localization performance is evaluated as a function of the IoU of the input proposals, and the detection performance as a function of IoU threshold, as in COCO [20]. Note that, in Figure 1 (c), each bounding box regressor performs best for examples of IoU close to the threshold that the detector was trained. This also holds for detection performance, up to overfitting. Figure 1 (d) shows that, the detector of $u = 0.5$ outperforms the detector of $u = 0.6$ for low IoU examples, underperforming it at higher IoU levels. In general, a detector optimized at a single IoU level is not necessarily optimal at other levels. These observations suggest that higher quality detection requires a closer quality *match* between the detector and the hypotheses that it processes. In general, a detector can only have high quality if presented with high quality proposals.

However, to produce a high quality detector, it does not suffice to simply increase u during training. In fact, as seen for the detector of $u = 0.7$ of Figure 1 (d), this can degrade detection performance. The problem is that the distribution of hypotheses out of a proposal detector is usually heavily imbalanced towards low quality. In general, forcing larger IoU thresholds leads to an exponentially smaller numbers of positive training samples. This is particularly problematic for neural networks, which are known to be very example intensive, and makes the “high u ” training strategy quite prone to overfitting. Another difficulty is the mismatch between the quality of the detector and that of the testing hypotheses at inference. As shown in Figure 1, high quality detectors are only necessarily optimal for high quality hypotheses. The detection could be suboptimal when they are asked to work on the hypotheses of other quality levels.

In this paper, we propose a new detector architecture, Cascade R-CNN, that addresses these problems. It is a multi-stage extension of the R-CNN, where detector stages deeper into the cascade are sequentially more selective against close false positives. The cascade of R-CNN stages are trained sequentially, using the output of one stage to train the next. This is motivated by the observation that the output IoU of a regressor is almost invariably better than the input IoU. This observation can be made in Figure 1

(c), where all plots are above the gray line. It suggests that the output of a detector trained with a certain IoU threshold is a good distribution to train the detector of the next higher IoU threshold. This is similar to *bootstrapping* methods commonly used to assemble datasets in object detection literature [31, 8]. The main difference is that the re-sampling procedure of the Cascade R-CNN does not aim to mine hard negatives. Instead, by adjusting bounding boxes, each stage aims to find a good set of close false positives for training the next stage. When operating in this manner, a sequence of detectors adapted to increasingly higher IoUs can beat the overfitting problem, and thus be effectively trained. At inference, the same cascade procedure is applied. The progressively improved hypotheses are better matched to the increasing detector quality at each stage. This enables higher detection accuracies, as suggested by Figure 1 (c) and (d).

The Cascade R-CNN is quite simple to implement and trained end-to-end. Our results show that a vanilla implementation, without any bells and whistles, surpasses all previous state-of-the-art *single-model* detectors by a large margin, on the challenging COCO detection task [20], especially under the higher quality evaluation metrics. In addition, the Cascade R-CNN can be built with any two-stage object detector based on the R-CNN framework. We have observed consistent gains (of 2~4 points), at a marginal increase in computation. This gain is independent of the strength of the baseline object detectors. We thus believe that this simple and effective detection architecture can be of interest for many object detection research efforts.

2. Related Work

Due to the success of the R-CNN [12] architecture, the two-stage formulation of the detection problems, by combining a proposal detector and a region-wise classifier has become predominant in the recent past. To reduce redundant CNN computations in the R-CNN, the SPP-Net [15] and Fast-RCNN [11] introduced the idea of region-wise feature extraction, significantly speeding up the overall detector. Later, the Faster-RCNN [27] achieved further speed-up by introducing a Region Proposal Network (RPN). This architecture has become a leading object detection framework. Some more recent works have extended it to address various problems of detail. For example, the R-FCN [4] proposed efficient region-wise fully convolutions without accuracy loss, to avoid the heavy region-wise CNN computations of the Faster-RCNN; while the MS-CNN [1] and FPN [21] detect proposals at multiple output layers, so as to alleviate the scale mismatch between the RPN receptive fields and actual object size, for high-recall proposal detection.

Alternatively, one-stage object detection architectures have also become popular, mostly due to their computa-

迄今为止，关于学习高质量物体检测器的问题研究甚少，这类检测器的输出应包含极少接近的误报，如图1(b)所示。其基本理念在于，单个检测器仅能针对单一质量水平达到最优。这在成本敏感学习文献[7, 24]中已有共识——优化接收者操作特征（ROC）曲线的不同点需要采用不同的损失函数。主要区别在于，我们关注的是在给定交并比（IoU）阈值下的优化，而非针对误报率进行优化。

该思想由图1(c)和(d)阐释，它们分别展示了使用IoU阈值为 $u=0.5, 0.6, 0.7$ 训练的三种检测器的定位与检测性能。定位性能根据输入建议框的IoU值进行评估，检测性能则依据IoU阈值的变化进行衡量，评估方式与COCO[20]一致。值得注意的是，在图1(c)中，每个边界框回归器对接近其训练阈值的IoU样本表现最佳。这一规律同样适用于检测性能，直至出现过拟合现象。图1(d)表明，在低IoU样本上，阈值为 $u=0.5$ 的检测器优于阈值为 $u=0.6$ 的检测器，但在高IoU水平上表现较弱。总体而言，针对单一IoU水平优化的检测器在其他水平上未必最优。这些观察结果表明，更高质量的检测需要检测器与其处理的假设之间具有更紧密的质量 $match$ 匹配。通常，检测器只有在接收高质量建议框时才能实现高质量检测。

然而，要生成高质量的检测器，仅仅在训练过程中提高 u 是不够的。实际上，如图1(d)中 $u=0.7$ 的检测器所示，这样做反而可能降低检测性能。问题在于，从候选检测器生成的假设分布通常严重偏向低质量。一般来说，强制提高IoU阈值会导致正训练样本数量呈指数级减少。这对神经网络尤其成问题，因为众所周知神经网络非常依赖大量样本，这使得“高 u ”训练策略极易过拟合。另一个困难在于检测器质量与推理时测试假设质量之间的不匹配。如图1所示，高质量的检测器仅对高质量假设必然最优。当要求它们处理其他质量水平的假设时，检测效果可能欠佳。

本文提出一种新的检测器架构——Cascade R-CNN，以解决上述问题。该架构是R-CNN的多阶段扩展，随着级联深度的增加，检测器阶段对近似假阳性样本的筛选能力逐步增强。级联的R-CNN阶段采用顺序训练方式，即利用前一阶段的输出来训练后一阶段。这种设计的动机源于我们观察到回归器的输出IoU几乎总是优于输入IoU，该现象可在图1中得到验证。

(c)，其中所有曲线均位于灰线之上。这表明，使用特定IoU阈值训练的检测器输出，是训练下一个更高IoU阈值检测器的良好分布。这与目标检测文献中常用的 $bootstrapping$ 方法类似[31, 8]。主要区别在于，Cascade R-CNN的重采样过程并非旨在挖掘困难负样本，而是通过调整边界框，使每个阶段都能为下一阶段的训练找到一组优质的接近假阳性样本。以这种方式操作时，一系列适应逐渐提高IoU阈值的检测器能够克服过拟合问题，从而得到有效训练。在推理时，采用相同的级联流程。逐步优化的假设与每一阶段不断提升的检测器质量更为匹配，如图1(c)和(d)所示，这有助于实现更高的检测精度。

Cascade R-CNN的实现和端到端训练都相当简单。我们的结果表明，在具有挑战性的COCO检测任务[20]中，一个未经修饰的普通实现，无需任何额外技巧，就以较大优势超越了所有先前最先进的 $single-model$ 检测器，尤其是在更高质量的评价指标下。此外，Cascade R-CNN可以基于R-CNN框架与任何两阶段目标检测器结合构建。我们观察到，在计算量略有增加的情况下，检测性能获得了稳定提升（提升幅度为2~4个百分点）。这种提升与基线目标检测器的性能强弱无关。因此，我们相信这种简单而有效的检测架构对许多目标检测研究工作都具有参考价值。

2. 相关工作

由于R-CNN[12]架构的成功，通过结合候选区域检测器和区域分类器来解决检测问题的两阶段方法，在近年来已成为主流。为了减少R-CNN中冗余的CNN计算，SPP-Net[15]和Fast-RCNN[11]引入了区域特征提取的思想，显著提升了整体检测器的速度。随后，Faster-RCNN[27]通过引入区域提议网络（RPN）进一步实现了加速。这一架构已成为领先的目标检测框架。近期的一些研究对其进行了扩展，以解决各种细节问题。例如，R-FCN[4]提出了高效的区域全卷积方法，在避免Faster-RCNN中繁重的区域CNN计算的同时保持了精度；而MS-CNN[1]和FPN[21]在多个输出层检测候选区域，以缓解RPN感受野与实际物体尺寸之间的尺度不匹配问题，从而实现高召回率的候选区域检测。

另外，单阶段目标检测架构也因其计算效率而变得流行起来，这主要归功于它们的计算——

tional efficiency. These architectures are close to the classic sliding window strategy [31, 8]. YOLO [26] outputs very sparse detection results by forwarding the input image once. When implemented with an efficient backbone network, it enables real time object detection with fair performance. SSD [23] detects objects in a way similar to the RPN [27], but uses multiple feature maps at different resolutions to cover objects at various scales. The main limitation of these architectures is that their accuracies are typically below that of two-stage detectors. Recently, RetinaNet [22] was proposed to address the extreme foreground-background class imbalance in dense object detection, achieving better results than state-of-the-art two-stage object detectors.

Some explorations in multi-stage object detection have also been proposed. The multi-region detector [9] introduced *iterative bounding box regression*, where a R-CNN is applied several times, to produce better bounding boxes. CRAFT [33] and AttractionNet [10] used a multi-stage procedure to generate accurate proposals, and forwarded them to a Fast-RCNN. [19, 25] embedded the classic cascade architecture of [31] in object detection networks. [3] iterated a detection and a segmentation task alternatively, for instance segmentation.

3. Object Detection

In this paper, we extend the two-stage architecture of the Faster-RCNN [27, 21], shown in Figure 3 (a). The first stage is a proposal sub-network (“H0”), applied to the entire image, to produce preliminary detection hypotheses, known as object proposals. In the second stage, these hypotheses are then processed by a region-of-interest detection sub-network (“H1”), denoted as detection head. A final classification score (“C”) and a bounding box (“B”) are assigned to each hypothesis. We focus on modeling a multi-stage detection sub-network, and adopt, but are not limited to, the RPN [27] for proposal detection.

3.1. Bounding Box Regression

A bounding box $\mathbf{b} = (b_x, b_y, b_w, b_h)$ contains the four coordinates of an image patch x . The task of bounding box regression is to regress a candidate bounding box $\mathbf{b}$ into a target bounding box $\mathbf{g}$, using a regressor $f(x, \mathbf{b})$. This is learned from a training sample $\{\mathbf{g}_i, \mathbf{b}_i\}$, so as to minimize the bounding box risk

$$\mathcal{R}_{loc}[f] = \sum_{i=1}^N L_{loc}(f(x_i, \mathbf{b}_i), \mathbf{g}_i), \quad (1)$$

where L_{loc} was a L_2 loss function in R-CNN [12], but updated to a smoothed L_1 loss function in Fast-RCNN [11]. To encourage a regression invariant to scale and location, L_{loc} operates on the distance vector $\Delta = (\delta_x, \delta_y, \delta_w, \delta_h)$

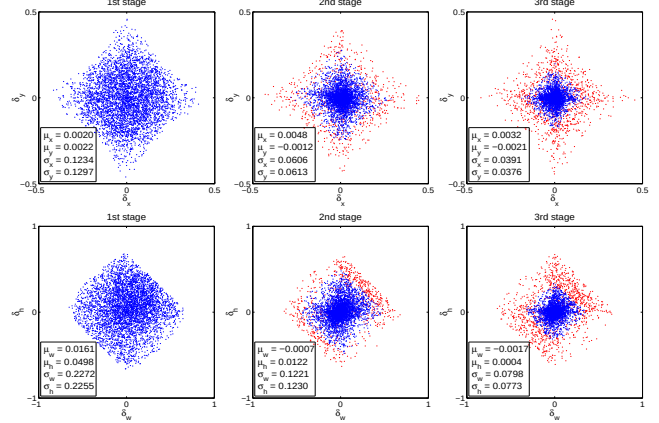


Figure 2. Sequential Δ distribution (without normalization) at different cascade stage. Red dots are outliers when using increasing IoU thresholds, and the statistics are obtained after outlier removal.

defined by

$$\begin{aligned} \delta_x &= (g_x - b_x)/b_w, & \delta_y &= (g_y - b_y)/b_h \\ \delta_w &= \log(g_w/b_w), & \delta_h &= \log(g_h/b_h). \end{aligned} \quad (2)$$

Since bounding box regression usually performs minor adjustments on b , the numerical values of (2) can be very small. Hence, the risk of (1) is usually much smaller than the classification risk. To improve the effectiveness of multi-task learning, Δ is usually normalized by its mean and variance, i.e. δ_x is replaced by $\delta'_x = (\delta_x - \mu_x)/\sigma_x$. This is widely used in the literature [27, 1, 4, 21, 14].

Some works [9, 10, 16] have argued that a single regression step of f is insufficient for accurate localization. Instead, f is applied iteratively, as a post-processing step

$$f'(x, \mathbf{b}) = f \circ f \circ \dots \circ f(x, \mathbf{b}), \quad (3)$$

to refine a bounding box $\mathbf{b}$. This is called *iterative bounding box regression*, denoted as *iterative BBox*. It can be implemented with the inference architecture of Figure 3 (b) where all heads are the same. This idea, however, ignores two problems. First, as shown in Figure 1, a regressor f trained at $u = 0.5$, is suboptimal for hypotheses of higher IoUs. It actually *degrades* bounding boxes of IoU larger than 0.85. Second, as shown in Figure 2, the distribution of bounding boxes changes significantly after each iteration. While the regressor is optimal for the initial distribution it can be quite suboptimal after that. Due to these problems, *iterative BBox* requires a fair amount of human engineering, in the form of proposal accumulation, box voting, etc. [9, 10, 16], and has somewhat unreliable gains. Usually, there is no benefit beyond applying f twice.

3.2. Classification

The classifier is a function $h(x)$ that assigns an image patch x to one of $M + 1$ classes, where class 0 contains

这些架构在计算效率上接近经典的滑动窗口策略[31, 8]。YOLO[26]通过对输入图像进行一次前向传播，输出非常稀疏的检测结果。当配合高效的主干网络实现时，它能以公平的性能实现实时目标检测。SSD[23]采用与RPN[27]类似的方式检测目标，但利用多分辨率特征图来覆盖不同尺度的目标。这类架构的主要局限在于其精度通常低于两阶段检测器。最近提出的RetinaNet[22]致力于解决密集目标检测中极端的前景-背景类别不平衡问题，取得了优于当前最先进两阶段目标检测器的结果。

在多阶段目标检测方面也提出了一些探索。多区域检测器[9]引入了 *iterative bounding box regression*，其中多次应用R-CNN以生成更优的边界框。CRAFT[33]和AttractionNet[10]采用多阶段流程生成精确的候选区域，并将其输入Fast-RCNN。[19, 25]将[31]的经典级联架构嵌入目标检测网络。[3]则通过交替迭代检测与分割任务来实现实例分割。

3. 目标检测

在本文中，我们扩展了Faster-RCNN[27, 21]的两阶段架构，如图3(a)所示。第一阶段是一个应用于整个图像的候选区域生成子网络（“H0”），用于产生初步的检测假设，即目标候选框。第二阶段中，这些假设由感兴趣区域检测子网络（“H1”）进行处理，该网络被称为检测头。每个假设都会被分配一个最终的分类分数（“C”）和一个边界框（“B”）。我们专注于建模多阶段检测子网络，并采用（但不限于）RPN[27]进行候选框检测。

3.1. 边界框回归

边界框 $\mathbf{b} = (b_x, b_y, b_w, b_h)$ 包含图像块 x 的四个坐标。边界框回归的任务是使用回归器 $f(x, \mathbf{b})$ ，将候选边界框 $\mathbf{b}$ 回归到目标边界框 $\mathbf{g}$ 。这是通过训练样本 $\{\mathbf{g}_i, \mathbf{b}_i\}$ 学习得到的，以最小化边界框风险。

$$\mathcal{R}_{loc}[f] = \sum_{i=1}^N L_{loc}(f(x_i, \mathbf{b}_i), \mathbf{g}_i), \quad (1)$$

其中 L_{loc} 在 R-CNN [12] 中是 L_2 损失函数，但在 Fast-RCNN [11] 中更新为平滑的 L_1 损失函数。为促使回归对尺度和位置具有不变性， L_{loc} 作用于距离向量 $\Delta = (\delta_x, \delta_y, \delta_w, \delta_h)$ 。

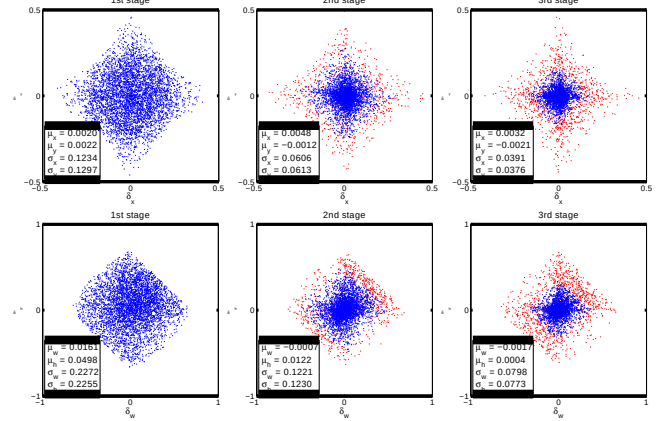


图2. 不同级联阶段下序列 Δ 的分布（未归一化）。红点为使用递增IoU阈值时的离群值，统计结果在剔除离群值后获得。

定义为

$$\begin{aligned} \delta_x &= (g_x - b_x)/b_w, & \delta_y &= (g_y - b_y)/b_h \\ \delta_w &= \log(g_w/b_w), & \delta_h &= \log(g_h/b_h). \end{aligned} \quad (2)$$

由于边界框回归通常对 $\{\mathbf{v}^*\}$ 进行微小的调整，(2)式的数值可能非常小。因此，(1)式的风险通常远小于分类风险。为了提高多任务学习的有效性， $\{\mathbf{v}^*\}$ 通常通过其均值和方差进行归一化，即用 $\{\mathbf{v}^*\}$ 替换 $\{\mathbf{v}^*\}$ 。这在文献[27, 1, 4, 21, 14]中被广泛使用。

一些研究[9, 10, 16]指出，仅通过单次回归步骤 f 难以实现精确定位。因此，通常采用迭代方式应用 f 作为后处理步骤。

$$f'(x, \mathbf{b}) = f \circ f \circ \dots \circ f(x, \mathbf{b}), \quad (3)$$

用于优化边界框 $\mathbf{b}$ 。这被称为 *iterative bounding box regression*，记作 *iterative BBox*。它可以通过图3(b)的推理架构实现，其中所有头结构相同。然而，这一思路忽略了两点问题。首先，如图1所示，在 $u=0.5$ 上训练的回归器 f 对于更高IoU的假设框并非最优选择——它实际上会 *degrades IoU* 大于0.85的边界框。其次，如图2所示，每次迭代后边界框的分布会发生显著变化。虽然回归器对初始分布是最优的，但在后续迭代中可能表现欠佳。由于这些问题，*iterative BBox* 需要大量人工设计（如候选框累积、边界框投票等[9,10,16]），且效果提升并不稳定。通常，重复应用 f 超过两次并不会带来额外收益。

3.2. 分类

分类器是一个函数 $h(x)$ ，它将图像块 x 分配给 $M + 1$ 个类别中的一类，其中类别 0 包含

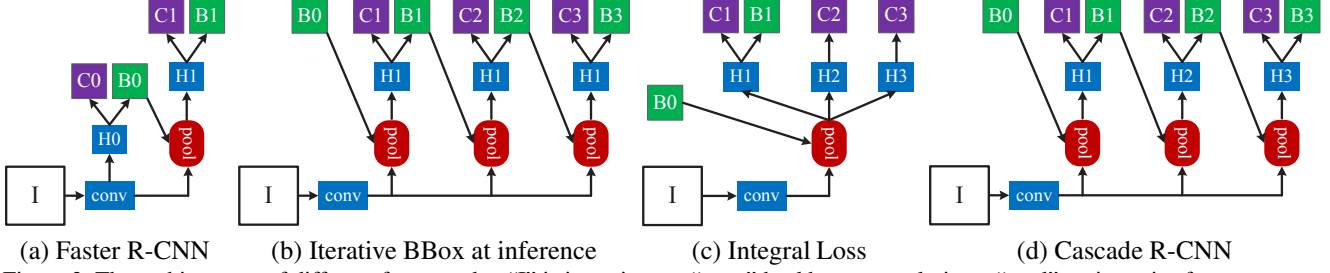


Figure 3. The architectures of different frameworks. “I” is input image, “conv” backbone convolutions, “pool” region-wise feature extraction, “H” network head, “B” bounding box, and “C” classification. “B0” is proposals in all architectures.

background and the remaining the objects to detect. $h(x)$ is a $M + 1$ -dimensional estimate of the posterior distribution over classes, i.e. $h_k(x) = p(y = k|x)$, where y is the class label. Given a training set (x_i, y_i) , it is learned by minimizing a classification risk

$$\mathcal{R}_{cls}[h] = \sum_{i=1}^N L_{cls}(h(x_i), y_i), \quad (4)$$

where L_{cls} is the classic cross-entropy loss.

3.3. Detection Quality

Since a bounding box usually includes an object and some amount of background, it is difficult to determine if a detection is positive or negative. This is usually addressed by the IoU metric. If the IoU is above a threshold u , the patch is considered an example of the class. Thus, the class label of a hypothesis x is a function of u ,

$$y = \begin{cases} g_y, & IoU(x, g) \geq u \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where g_y is the class label of the ground truth object g . This IoU threshold u defines the quality of a detector.

Object detection is challenging because, no matter threshold, the detection setting is highly adversarial. When u is high, the positives contain less background, but it is difficult to assemble enough positive training examples. When u is low, a richer and more diversified positive training set is available, but the trained detector has little incentive to reject close false positives. In general, it is very difficult to ask a single classifier to perform uniformly well over all IoU levels. At inference, since the majority of the hypotheses produced by a proposal detector, e.g. RPN [27] or selective search [30], have low quality, the detector must be more discriminant for lower quality hypotheses. A standard compromise between these conflicting requirements is to settle on $u = 0.5$. This, however, is a relatively low threshold, leading to low quality detections that most humans consider close false positives, as shown in Figure 1 (a).

A naïve solution is to develop an ensemble of classifiers, with the architecture of Figure 3 (c), optimized with a loss

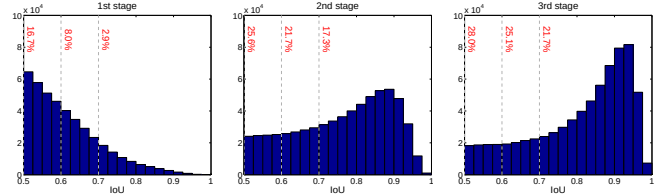


Figure 4. The IoU histogram of training samples. The distribution at 1st stage is the output of RPN. The red numbers are the positive percentage higher than the corresponding IoU threshold.

that targets various quality levels,

$$L_{cls}(h(x), y) = \sum_{u \in U} L_{cls}(h_u(x), y_u), \quad (6)$$

where U is a set of IoU thresholds. This is closely related to the *integral loss* of [34], in which $U = \{0.5, 0.55, \dots, 0.75\}$, designed to fit the evaluation metric of the COCO challenge. By definition, the classifiers need to be ensembled at inference. This solution fails to address the problem that the different losses of (6) operate on different numbers of positives. As shown in the first figure of Figure 4, the set of positive samples decreases quickly with u . This is particularly problematic because the high quality classifiers are prone to overfitting. In addition, those high quality classifiers are required to process proposals of overwhelming low quality at inference, for which they are not optimized. Due to all this, the ensemble of (6) fails to achieve higher accuracy at most quality levels, and the architecture has very little gain over that of Figure 3 (a).

4. Cascade R-CNN

In this section we introduce the proposed Cascade R-CNN object detection architecture of Figure 3 (d).

4.1. Cascaded Bounding Box Regression

As seen in Figure 1 (c), it is very difficult to ask a single regressor to perform perfectly uniformly at all quality levels. The difficult regression task can be decomposed into a sequence of simpler steps, inspired by the works of cascade pose regression [6] and face alignment [2, 32]. In the

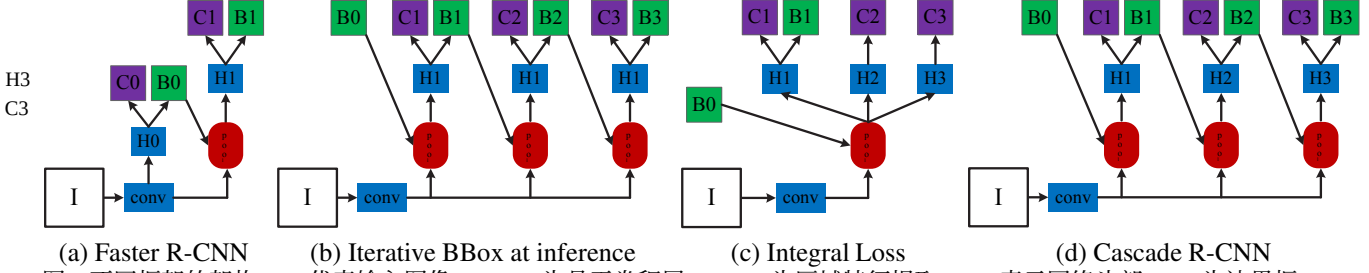


图3. 不同框架的架构。"I"代表输入图像，"conv"为骨干卷积层，"pool"为区域特征提取，"H"表示网络头部，"B"为边界框，"C"为分类器。"B0"是所有架构中的候选区域。

背景以及待检测的其余对象。 $h(x)$ 是类别后验分布的一维估计 $M+$ ，即 $h_k(x) = p(y = k|x)$ ，其中 y 为类别标签。在给定训练集 (x_i, y_i) 的情况下，通过最小化分类风险进行学习。

$$\mathcal{R}_{cls}[h] = \sum_{i=1}^N L_{cls}(h(x_i), y_i), \quad (4)$$

其中 L_{cls} 是经典的交叉熵损失。

3.3. 检测质量

由于边界框通常包含物体和一定量的背景，因此很难判断检测结果是阳性还是阴性。这一问题通常通过IoU指标来解决。如果IoU高于阈值 u ，该图像块就被视为该类别的正样本。因此，假设 x 的类别标签是 u 的函数，

$$y = \begin{cases} g_y, & IoU(x, g) \geq u \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

其中 g_y 是真实目标 g 的类别标签。该 IoU 阈值 u 定义了检测器的质量。

目标检测具有挑战性，因为无论阈值如何设定，检测环境都处于高度对抗状态。当 u 较高时，正样本包含的背景较少，但难以收集足够的正训练样本。当 u 较低时，可以获得更丰富、更多样化的正训练集，但训练出的检测器几乎没有动力去排除接近的误报。总的来说，要求单个分类器在所有IoU级别上均表现优异是非常困难的。在推理阶段，由于提案检测器（如RPN[27]或选择性搜索[30]）产生的大部分假设质量较低，检测器必须对低质量假设更具判别力。这些冲突需求之间的标准折衷方案是将阈值设定为 $u = 0.5$ 。然而，这是一个相对较低的阈值，会导致产生大多数人类视为接近误报的低质量检测结果，如图1(a)所示。

一种天真的解决方案是开发一个分类器集成，采用图3(c)的架构，并通过损失函数进行优化。

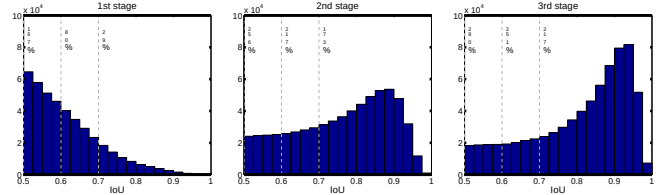


图4. 训练样本的IoU直方图。第一阶段的分布是RPN的输出结果。红色数字表示高于对应IoU阈值的正样本百分比。

针对不同质量水平，

$$L_{cls}(h(x), y) = \sum_{u \in U} L_{cls}(h_u(x), y_u), \quad (6)$$

其中 U 是一组IoU阈值。这与[34]中的 *integral loss* 密切相关，其中 $U = \{0.5, 0.55, \dots, 0.75\}$ 的设计旨在匹配COCO挑战赛的评估指标。根据定义，分类器需要在推理时进行集成。这一方案未能解决(6)式中不同损失函数作用于不同数量正样本的问题。如图4首图所示，正样本集合随 u 增加而迅速减少。这尤其成问题，因为高质量分类器容易过拟合。此外，这些高质量分类器在推理时需处理大量低质量提案，而它们并未针对此类数据优化。基于所有这些原因，(6)式的集成方案在多数质量级别上未能实现更高精度，且该架构相比图3(a)几乎没有任何提升。

4. 级联R-CNN

在本节中，我们介绍图3(d)所示的级联R-CNN目标检测架构。

4.1. 级联边界框回归

如图1(c)所示，要求单个回归器在所有质量级别上完全均匀地完美执行是非常困难的。受级联姿态回归[6]和人脸对齐[2, 32]研究的启发，这一困难回归任务可分解为一系列更简单的步骤。在

Cascade R-CNN, it is framed as a cascaded regression problem, with the architecture of Figure 3 (d). This relies on a cascade of *specialized* regressors

$$f(x, \mathbf{b}) = f_T \circ f_{T-1} \circ \dots \circ f_1(x, \mathbf{b}), \quad (7)$$

where T is the total number of cascade stages. Note that each regressor f_t in the cascade is optimized *w.r.t.* the sample distribution $\{\mathbf{b}^t\}$ arriving at the corresponding stage, instead of the initial distribution of $\{\mathbf{b}^1\}$. This cascade improves hypotheses progressively.

It differs from the *iterative BBox* architecture of Figure 3 (b) in several ways. First, while *iterative BBox* is a post-processing procedure used to improve bounding boxes, cascaded regression is a *resampling* procedure that changes the distribution of hypotheses to be processed by the different stages. Second, because it is used at both training and inference, there is no discrepancy between training and inference distributions. Third, the multiple specialized regressors $\{f_T, f_{T-1}, \dots, f_1\}$ are optimized for the *resampled distributions* of the different stages. This opposes to the single f of (3), which is only optimal for the initial distribution. These differences enable more precise localization than *iterative BBox*, with no further human engineering.

As discussed in Section 3.1, $\Delta = (\delta_x, \delta_y, \delta_w, \delta_h)$ in (2) needs to be normalized by its mean and variance for effective multi-task learning. After each regression stage, these statistics will evolve sequentially, as displayed in Figure 2. At training, the corresponding statistics are used to normalize Δ at each stage.

4.2. Cascaded Detection

As shown in the left of Figure 4, the distribution of the initial hypotheses, e.g. RPN proposals, is heavily tilted towards low quality. This inevitably induces ineffective learning of higher quality classifiers. The Cascade R-CNN addresses the problem by relying on cascade regression as a *resampling mechanism*. This is motivated by the fact that in Figure 1 (c) all curves are above the diagonal gray line, i.e. a bounding box regressor trained for a certain u tends to produce bounding boxes of *higher* IoU. Hence, starting from a set of examples $(x_i, \mathbf{b}_i)$, cascade regression successively resamples an example distribution $(x'_i, \mathbf{b}'_i)$ of higher IoU. In this manner, it is possible to keep the set of positive examples of the successive stages at a roughly *constant* size, even when the detector quality (IoU threshold) is *increased*. This is illustrated in Figure 4, where the distribution tilts more heavily towards high quality examples after each resampling step. Two consequences ensue. First, there is no overfitting, since examples are plentiful at all levels. Second, the detectors of the deeper stages are optimized for higher IoU thresholds. Note that, some outliers are sequentially removed by increasing IoU thresholds, as illustrated

in Figure 2, enabling a better trained sequence of specialized detectors.

At each stage t , the R-CNN includes a classifier h_t and a regressor f_t optimized for IoU threshold u^t , where $u^t > u^{t-1}$. This is guaranteed by minimizing the loss

$$L(x^t, g) = L_{cls}(h_t(x^t), y^t) + \lambda[y^t \geq 1]L_{loc}(f_t(x^t, \mathbf{b}^t), \mathbf{g}), \quad (8)$$

where $\mathbf{b}^t = f_{t-1}(x^{t-1}, \mathbf{b}^{t-1})$, g is the ground truth object for x^t , $\lambda = 1$ the trade-off coefficient, $[\cdot]$ the indicator function, and y^t is the label of x^t given u^t by (5). Unlike the *integral loss* of (6), this guarantees a sequence of effectively trained detectors of increasing quality. At inference, the quality of the hypotheses is sequentially improved, by applications of the same cascade procedure, and higher quality detectors are only required to operate on higher quality hypotheses. This enables high quality object detection, as suggested by Figure 1 (c) and (d).

5. Experimental Results

The Cascade R-CNN was evaluated on MS-COCO 2017 [20], which contains $\sim 118k$ images for training, 5k for validation (`val`) and $\sim 20k$ for testing without provided annotations (`test-dev`). The COCO-style Average Precision (AP) averages AP across IoU thresholds from 0.5 to 0.95 with an interval of 0.05. These evaluation metrics measure the detection performance of various qualities. All models were trained on COCO training set, and evaluated on `val` set. Final results were also reported on `test-dev` set.

5.1. Implementation Details

All regressors are class agnostic for simplicity. All cascade detection stages in Cascade R-CNN have the same architecture, which is the head of the baseline detection network. In total, Cascade R-CNN have four stages, one RPN and three for detection with $U = \{0.5, 0.6, 0.7\}$, unless otherwise noted. The sampling of the first detection stage follows [11, 27]. In the following stages, resampling is implemented by simply using the regressed outputs from the previous stage, as in Section 4.2. No data augmentation was used except standard horizontal image flipping. Inference was performed on a single image scale, with no further bells and whistles. All baseline detectors were reimplemented with Caffe [18], on the same codebase for fair comparison.

5.1.1 Baseline Networks

To test the versatility of the Cascade R-CNN, experiments were performed with three popular baseline detectors: Faster-RCNN with backbone VGG-Net [29], R-FCN [4] and FPN [21] with ResNet backbone [16]. These baselines have a wide range of detection performances. Unless noted, their default settings were used. End-to-end training was used instead of multi-step training.

Cascade R-CNN被构建为一个级联回归问题，其架构如图3 (d) 所示。该方法依赖于一个由*specialized*回归器组成的级联结构。

$$f(x, \mathbf{b}) = f_T \circ f_{T-1} \circ \dots \circ f_1(x, \mathbf{b}), \quad (7)$$

其中 T 是级联阶段的总数。注意，级联中的每个回归器 f_t 都是针对到达对应阶段的样本分布 $\{\mathbf{b}^t\}$ 进行优化的，而非针对初始分布 $\{\mathbf{b}^1\}$ 。这种级联结构能逐步改进假设。

它与图3(b)中的*iterative BBox*架构在几个方面有所不同。首先，*iterative BBox*是用于改进边界框的后处理过程，而级联回归是一种*resampling*过程，它改变了不同阶段待处理假设的分布。其次，由于它在训练和推理中都被使用，因此训练和推理分布之间没有差异。第三，多个专用回归器 $\{f_T, f_{T-1}, \dots, f_1\}$ 针对不同阶段的*resampled distributions*进行了优化。这与(3)中的单一 f 形成对比，后者仅对初始分布最优。这些差异使得定位比*iterative BBox*更精确，且无需额外的人工设计。

如第3.1节所述，为进行有效的多任务学习，(2)式中的 $\{\mathbf{v}^*\}$ 需按其均值与方差进行归一化。在每个回归阶段后，这些统计量将如图2所示依次更新。训练时，各阶段使用对应的统计量对 $\{\mathbf{v}^*\}$ 进行归一化处理。

4.2. 级联检测

如图4左侧所示，初始假设（例如RPN提议）的分布严重偏向低质量区域。这不可避免地导致高质量分类器的学习效率低下。Cascade R-CNN通过依赖级联回归作为*resampling mechanism*来解决这一问题。其动机在于图1(c)中所有曲线均高于对角灰色线，即针对特定 u 训练的边界框回归器倾向于生成*higher* IoU的边界框。因此，从一组样本 $(x_i, \mathbf{b}_i)$ 出发，级联回归逐步重采样出更高IoU的样本分布 $(x'_i, \mathbf{b}'_i)$ 。通过这种方式，即使检测器质量（IoU阈值）*increased*，仍能保持后续阶段正样本集的规模大致*constant*。图4展示了这一过程：每次重采样后，分布更显著地向高质量样本倾斜。由此产生两个结果：首先，由于各层级样本充足，避免了过拟合；其次，更深阶段的检测器针对更高IoU阈值进行了优化。需注意，如示意图所示，部分异常值会通过提升IoU阈值被逐步剔除。

在图2中，启用了训练更佳的专业化检测器序列。

在每个阶段 t ，R-CNN都包含一个针对IoU阈值 u^t 优化的分类器 h_t 和回归器 f_t ，其中 $u^t > u^{t-1}$ 。这是通过最小化损失来保证的

$$L(x^t, g) = L_{cls}(h_t(x^t), y^t) + \lambda[y^t \geq 1]L_{loc}(f_t(x^t, \mathbf{b}^t), \mathbf{g}), \quad (8)$$

其中 $\mathbf{b}^t = f_{t-1}(x^{t-1}, \mathbf{b}^{t-1})$ 、 g 是 x^t 的真实标注对象， λ 为权衡系数， $[\cdot]$ 是指示函数， y^t 是通过公式(5)在给定 u^t 条件下 x^t 的标签。与公式(6)的*integral loss*不同，这种方法保证了能训练出一系列质量逐步提升的有效检测器。在推理阶段，通过应用相同的级联流程，假设的质量被逐步提升，而更高质量的检测器仅需处理更高质量的假设。如图1 (c) 和 (d) 所示，这实现了高质量的物体检测。

5. 实验结果

Cascade R-CNN在MS-COCO 2017数据集[20]上进行了评估，该数据集包含~11.8万张训练图像、5千张验证集图像以及~2万张未提供标注的测试集图像（test-dev）。采用COCO风格的平均精度（AP）作为评估指标，其计算方式是在0.5至0.95的IoU阈值范围内以0.05为间隔取AP的平均值。这些评估指标能衡量不同质量等级的检测性能。所有模型均在COCO训练集上训练，并在验证集上评估，最终结果也在test-dev集上进行了报告。

5.1. 实现细节

为简化起见，所有回归器均为类别无关设计。Cascade R-CNN中的级联检测阶段均采用相同架构，即基线检测网络的头部结构。整个系统共包含四个阶段：一个区域提议网络和三个检测阶段（IoU阈值分别设为 $U = \{0.5, 0.6, 0.7\}$ ），除非另有说明。首阶段检测采样遵循[11, 27]方法，后续阶段通过直接使用前一阶段的回归输出进行重采样（如第4.2节所述）。除标准水平图像翻转外未使用数据增强。推理过程在单图像尺度下进行，未添加其他优化技巧。所有基线检测器均基于Caffe[18]在同一代码库中重新实现，以确保公平比较。

5.1.1 基线网络

为了测试Cascade R-CNN的通用性，我们使用三种流行的基线检测器进行了实验：基于VGG-Net[29]的Faster-RCNN、R-FCN[4]以及基于ResNet[16]的FPN[21]。这些基线模型具有广泛的检测性能范围。除非特别说明，实验均采用其默认设置，并使用端到端训练替代多步训练。

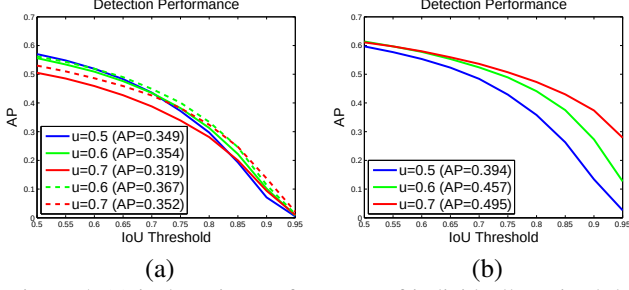


Figure 5. (a) is detection performance of individually trained detectors, with their own proposals (solid curves) or Cascade R-CNN stage proposals (dashed curves), and (b) is by adding ground truth to the proposal set.

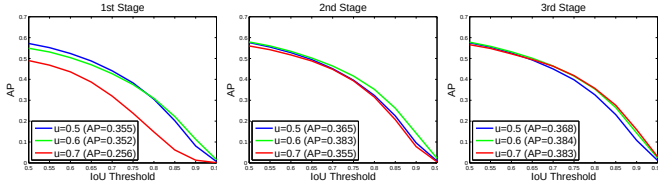


Figure 6. The detection performance of all Cascade R-CNN detectors at all cascade stages.

Faster-RCNN: The network head has two fully connected layers. To reduce parameters, we used [13] to prune less important connections. 2048 units were retained per fully connected layer and dropout layers were removed. Training started with a learning rate of 0.002, reduced by a factor of 10 at 60k and 90k iterations, and stopped at 100k iterations, on 2 synchronized GPUs, each holding 4 images per iteration. 128 RoIs were used per image.

R-FCN: R-FCN adds a convolutional, a bounding box regression, and a classification layer to the ResNet. All heads of the Cascade R-CNN have this structure. Online hard negative mining [28] was not used. Training started with a learning rate of 0.003, which was decreased by a factor of 10 at 160k and 240k iterations, and stopped at 280k iterations, on 4 synchronized GPUs, each holding one image per iteration. 256 RoIs were used per image.

FPN: Since no source code is publicly available yet for FPN, our implementation details could be different. RoIAlign [14] was used for a stronger baseline. This is denoted as FPN+ and was used in all ablation studies. As usual, ResNet-50 was used for ablation studies, and ResNet-101 for final detection. Training used a learning rate of 0.005 for 120k iterations and 0.0005 for the next 60k iterations, on 8 synchronized GPUs, each holding one image per iteration. 256 RoIs were used per image.

5.2. Quality Mismatch

Figure 5 (a) shows the AP curves of three individually trained detectors of increasing IoU thresholds of $U =$

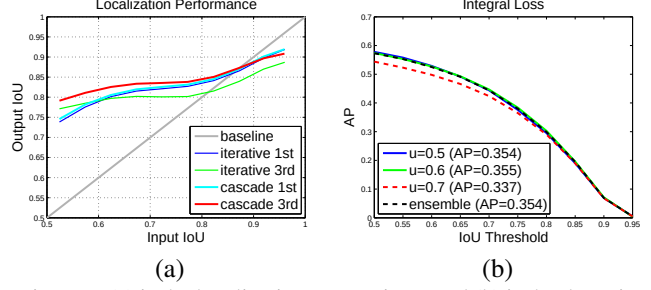


Figure 7. (a) is the localization comparison, and (b) is the detection performance of individual classifiers in the *integral loss* detector.

$\{0.5, 0.6, 0.7\}$. The detector of $u = 0.5$ outperforms the detector of $u = 0.6$ at low IoU levels, but underperforms it at higher levels. However, the detector of $u = 0.7$ underperforms the other two. To understand why this happens, we changed the quality of the proposals at inference. Figure 5 (b) shows the results obtained when ground truth bounding boxes were added to the set of proposals. While all detectors improve, the detector of $u = 0.7$ has the largest gains, achieving the best performance at almost all IoU levels. These results suggest two conclusions. First, $u = 0.5$ is not a good choice for precise detection, simply more robust to low quality proposals. Second, highly precise detection requires hypotheses that match the detector quality. Next, the original detector proposals were replaced by the Cascade R-CNN proposals of higher quality ($u = 0.6$ and $u = 0.7$ used the 2nd and 3rd stage proposals, respectively). Figure 5 (a) also suggests that the performance of the two detectors is significantly improved when the testing proposals closer match the detector quality.

Testing all Cascade R-CNN detectors at all cascade stages produced similar observations. Figure 6 shows that each detector was improved when used more precise hypotheses, while higher quality detector had larger gain. For example, the detector of $u = 0.7$ performed poorly for the low quality proposals of the 1st stage, but much better for the more precise hypotheses available at the deeper cascade stages. In addition, the jointly trained detectors of Figure 6 outperformed the individually trained detectors of Figure 5 (a), even when the same proposals were used. This indicates that the detectors are better trained within the Cascade R-CNN framework.

5.3. Comparison with *Iterative BBox* and *Integral Loss*

In this section, we compare the Cascade R-CNN to *iterative BBox* and the *integral loss* detector. *Iterative BBox* was implemented by applying the FPN+ baseline iteratively, three times. The *integral loss* detector has the same number of classification heads as the Cascade R-CNN, with $U = \{0.5, 0.6, 0.7\}$.

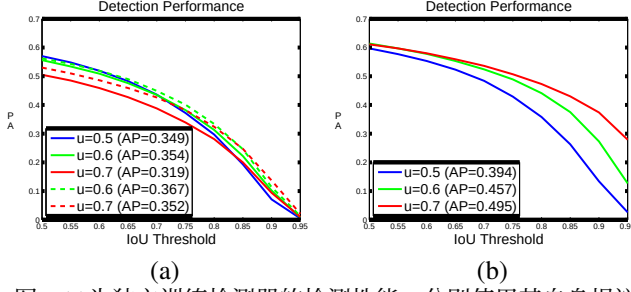


图5. (a)为独立训练检测器的检测性能，分别使用其自身提议（实线曲线）或级联R-CNN阶段提议（虚线曲线），(b)为在提议集中加入真实标注后的检测性能。

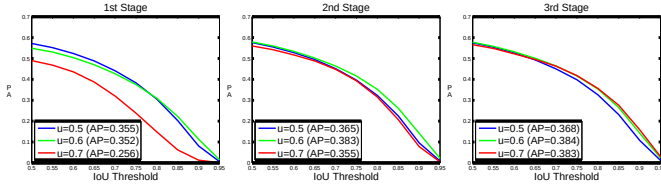


图6. 所有Cascade R-CNN检测器在各级联阶段的检测性能。

Faster-RCNN: 网络头部包含两个全连接层。为减少参数量，我们采用[13]方法剪枝了重要性较低的连接。每个全连接层保留2048个单元，并移除了dropout层。训练初始学习率为0.002，在6万次和9万次迭代时学习率降至十分之一，于10万次迭代时终止。训练使用2台同步GPU，每次迭代每台GPU处理4张图像。每张图像使用128个候选区域。

R-FCN: R-FCN在ResNet基础上增加了一个卷积层、一个边界框回归层和一个分类层。Cascade R-CNN的所有检测头均采用此结构。训练中未使用在线难负样本挖掘[28]。初始学习率为0.003，在16万次和24万次迭代时学习率下降10倍，训练在4台同步GPU上进行至28万次迭代停止，每次迭代每张GPU处理一张图像。每张图像使用256个候选区域。

FPN: 由于目前尚无公开的FPN源代码，我们的实现细节可能有所不同。为构建更强的基线模型，我们采用了RoIAlign[14]方法。该版本标记为FPN+，并用于所有消融实验。按照惯例，消融研究使用ResNet-50骨干网络，最终检测则采用ResNet-101。训练使用8个同步GPU，每轮迭代每张GPU处理一张图像，前12万次迭代学习率为0.005，后续6万次迭代降至0.0005。每张图像使用256个RoI。

5.2. 质量不匹配

图5 (a) 展示了三个独立训练的检测器在不同IoU阈值下的AP曲线，阈值分别为 $U =$

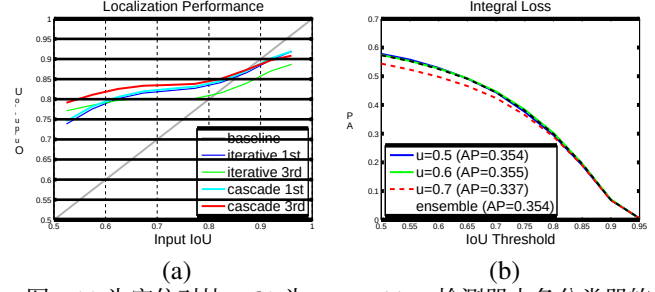


图7. (a) 为定位对比，(b) 为 *integral loss* 检测器中各分类器的检测性能。

$\{0.5, 0.6, 0.7\}$ 。 $u = 0.5$ 的检测器在低IoU水平上优于 $u = 0.6$ 的检测器，但在更高水平上表现较差。然而， $u = 0.7$ 的检测器表现不如另外两者。为了理解其原因，我们在推理时改变了提议的质量。图5(b)展示了将真实边界框加入提议集后得到的结果。虽然所有检测器都有所提升，但 $u = 0.7$ 的检测器增益最大，在几乎所有IoU水平上都达到了最佳性能。这些结果提示了两个结论：第一， $u = 0.5$ 并非精确检测的理想选择，仅对低质量提议更具鲁棒性；第二，高精度检测需要与检测器质量匹配的假设。接着，原始检测器提议被更高质量的Cascade R-CNN提议所取代（ $u = 0.6$ 和 $u = 0.7$ 分别使用了第2和第3阶段的提议）。图5(a)同样表明，当测试提议更贴近检测器质量时，两个检测器的性能均得到显著提升。

在所有级联阶段测试所有Cascade R-CNN检测器均得出相似观察结果。图6显示，当使用更精确的假设时，每个检测器性能均得到提升，而更高质量的检测器增益更大。例如， $u = 0.7$ 的检测器在第一阶段的低质量候选框上表现不佳，但在更深级联阶段提供的更精确假设上表现显著改善。此外，即使使用相同候选框，图6中联合训练的检测器也优于图5 (a) 中单独训练的检测器。这表明在Cascade R-CNN框架内训练的检测器效果更优。

5.3. 与 *Iterative BBox* 和 *Integral Loss* 的对比

在本节中，我们将Cascade R-CNN与*iterative BBox*以及*integral loss*检测器进行比较。*Iterative BBox*是通过将FPN+基线进行三次迭代应用实现的。*integral loss*检测器具有与Cascade R-CNN相同数量的分类头，其参数为 $U = \{0.5, 0.6, 0.7\}$ 。

	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
FPN+ baseline	34.9	57.0	51.9	43.6	29.7	7.1
<i>Iterative BBox</i>	35.4	57.2	52.1	44.2	30.4	8.1
<i>Integral Loss</i>	35.4	57.3	52.5	44.4	29.9	6.9
Cascade R-CNN	38.9	57.8	53.4	46.9	35.8	15.8

Table 1. The comparison with *iterative BBox* and *integral loss*.

test stage	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
1	35.5	57.2	52.4	44.1	30.5	8.1
2	38.3	57.9	53.4	46.4	35.2	14.2
3	38.3	56.6	52.2	46.3	35.7	15.9
$\overline{1 \sim 2}$	38.5	58.2	53.8	46.7	35.0	14.0
$\overline{1 \sim 3}$	38.9	57.8	53.4	46.9	35.8	15.8
FPN+ baseline	34.9	57.0	51.9	43.6	29.7	7.1

Table 2. The stage performance of Cascade R-CNN. $\overline{1 \sim 3}$ indicates the ensemble of three classifiers on the 3rd stage proposals.

Localization: The localization performances of cascade regression and *iterative BBox* are compared in Figure 7 (a). The use of a single regressor degrades localization for hypotheses of high IoU. This effect accumulates when the regressor is applied iteratively, as in *iterative BBox*, and performance actually drops. Note the very poor performance of *iterative BBox* after 3 iterations. On the contrary, the cascade regressor has better performance at later stages, outperforming *iterative BBox* at almost all IoU levels.

Integral Loss: The detection performances of all classifiers in the *integral loss* detector, sharing a single regressor, are shown in Figure 7 (b). The classifier of $u = 0.6$ is the best at all IoU levels, while the classifier of $u = 0.7$ is the worst. The ensemble of all classifiers shows no visible gain.

Table 1 shows, both *iterative BBox* and *integral loss* detector improve the baseline detector marginally. The cascade R-CNN has the best performance for all evaluation metrics. The gains are mild for low IoU thresholds but significant for the higher ones.

5.4. Ablation Experiments

Ablation experiments were also performed.

Stage-wise Comparison: Table 2 summarizes stage performance. The 1st stage already outperforms the baseline detector, due to the benefits of multi-stage multi-task learning. The 2nd stage improves performance substantially, and the 3rd is equivalent to the 2nd. This differs from the *integral loss* detector, where the higher IOU classifier is relatively weak. While the former (later) stage is better at low (high) IoU metrics, the ensemble of all classifiers is the best overall.

IoU Thresholds: A preliminary Cascade R-CNN was trained using the same IoU threshold $u = 0.5$ for all heads. In this case, the stages differ only in the hypotheses they

IoU $\uparrow$	stat	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
		36.8	57.8	52.9	45.4	32.0	10.7
✓		38.5	58.4	54.1	47.1	35.0	13.1
	✓	37.5	57.8	53.1	45.5	33.3	13.1
✓	✓	38.9	57.8	53.4	46.9	35.8	15.8

Table 3. The ablation experiments. “IoU $\uparrow$ ” means increasing IoU thresholds, and “stat” exploiting sequential regression statistics.

# stages	test stage	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
1	1	34.9	57.0	51.9	43.6	29.7	7.1
2	$\overline{1 \sim 2}$	38.2	58.0	53.6	46.7	34.6	13.6
3	$\overline{1 \sim 3}$	38.9	57.8	53.4	46.9	35.8	15.8
4	$\overline{1 \sim 3}$	38.9	57.4	53.2	46.8	36.0	16.0
4	$\overline{1 \sim 4}$	38.6	57.2	52.8	46.2	35.5	16.3

Table 4. The impact of the number of stages in Cascade R-CNN.

receive. Each stage is trained with the corresponding hypotheses, i.e. accounting for the distributions of Figure 2. The first row of Table 3 shows that the cascade improves on the baseline detector. This suggests the importance of optimizing stages for the corresponding sample distributions. The second row shows that, by increasing the stage threshold u , the detector can be made more selective against close false positives and *specialized* for more precise hypotheses, leading to additional gains. This supports the conclusions of Section 4.2.

Regression Statistics: Exploiting the progressively updated regression statistics, of Figure 2, helps the effective multi-task learning of classification and regression. Its benefit is noted by comparing the models with/without it in Table 3. The learning is not sensitive to these statistics.

Number of Stages: The impact of the number of stages is summarized in Table 4. Adding a second detection stage significantly improves the baseline detector. Three detection stages still produce non-trivial improvement, but the addition of a 4th stage ($u = 0.75$) led to a slight performance decrease. Note, however, that while the overall AP performance degrades, the four-stage cascade has the best performance for high IoU levels. The three-stage cascade achieves the best trade-off.

5.5. Comparison with the state-of-the-art

The Cascade R-CNN, based on FPN+ and ResNet-101 backbone, is compared to state-of-the-art *single-model* object detectors in Table 5. The settings are as described in Section 5.1.1, but a total of 280k training iterations were run and the learning rate dropped at 160k and 240k iterations. The number of RoIs was also increased to 512. The first group of detectors on Table 5 are one-stage detectors, the second group two-stage, and the last group multi-stage (3-stages+RPN for the Cascade R-CNN). All the compared state-of-the-art detectors were trained with $u = 0.5$. It is

	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
FPN+ baseline	34.9	57.0	51.9	43.6	29.7	7.1
<i>Iterative BBox</i>	35.4	57.2	52.1	44.2	30.4	8.1
<i>Integral Loss</i>	35.4	57.3	52.5	44.4	29.9	6.9
Cascade R-CNN	38.9	57.8	53.4	46.9	35.8	15.8

表1. 与*iterative BBox*和*integral loss*的对比。

test stage	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
1	35.5	57.2	52.4	44.1	30.5	8.1
2	38.3	57.9	53.4	46.4	35.2	14.2
3	38.3	56.6	52.2	46.3	35.7	15.9
1 ~ 2	38.5	58.2	53.8	46.7	35.0	14.0
1 ~ 3	38.9	57.8	53.4	46.9	35.8	15.8
FPN+ baseline	34.9	57.0	51.9	43.6	29.7	7.1

表2. Cascade R-CNN的阶段性能。1 ~ 3表示在第三阶段提议上三个分类器的集成。

定位：级联回归与*iterative BBox*的定位性能在图7(a)中进行比较。使用单一回归器会降低高IoU假设的定位精度。当回归器如*iterative BBox*中那样迭代应用时，这种影响会累积，性能实际上会下降。注意*iterative BBox*在3次迭代后表现极差。相反，级联回归器在后期阶段表现更好，在几乎所有IoU水平上都优于*iterative BBox*。

积分损失：*integral loss*检测器中所有分类器的检测性能，共享一个回归器，如图7(b)所示。 $u = 0.6$ 的分类器在所有IoU水平上表现最佳，而 $u = 0.7$ 的分类器表现最差。所有分类器的集成未显示出明显增益。

表1显示，*iterative BBox*和*integral loss*检测器均较基线检测器略有提升。级联R-CNN在所有评估指标上均取得最佳性能，其在低IoU阈值下提升幅度较小，但在较高阈值下改善显著。

5.4. 消融实验

还进行了消融实验。

阶段对比：表2总结了各阶段性能。由于多阶段多任务学习的优势，第一阶段已超越基线检测器。第二阶段性能显著提升，第三阶段与第二阶段持平。这与*integral loss*检测器不同，后者的高IoU分类器相对较弱。虽然前（后）期阶段在低（高）IoU指标上表现更优，但所有分类器的集成模型整体表现最佳。

IoU阈值：一个初步的Cascade R-CNN使用相同的IoU阈值 $u = 0.5$ 对所有头部进行训练。在这种情况下，各阶段仅在其处理的假设上有所不同。

IoU↑	stat	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
		36.8	57.8	52.9	45.4	32.0	10.7
✓		38.5	58.4	54.1	47.1	35.0	13.1
	✓	37.5	57.8	53.1	45.5	33.3	13.1
✓	✓	38.9	57.8	53.4	46.9	35.8	15.8

表3. 消融实验。"IoU↑"表示提高IoU阈值，"stat"表示利用序列回归统计。

# stages	test stage	AP	AP ₅₀	AP ₆₀	AP ₇₀	AP ₈₀	AP ₉₀
1	1	34.9	57.0	51.9	43.6	29.7	7.1
2	1 ~ 2	38.2	58.0	53.6	46.7	34.6	13.6
3	1 ~ 3	38.9	57.8	53.4	46.9	35.8	15.8
4	1 ~ 3	38.9	57.4	53.2	46.8	36.0	16.0
4	1 ~ 4	38.6	57.2	52.8	46.2	35.5	16.3

表4. Cascade R-CNN中阶段数量对性能的影响。

接收。每个阶段都使用相应的假设进行训练，即考虑到图2中的分布。表3的第一行显示，级联方法在基线检测器的基础上有所改进。这表明针对相应样本分布优化各阶段的重要性。第二行显示，通过提高阶段阈值 u ，检测器可以对接近的误报更具选择性，并*specialized*以获取更精确的假设，从而带来额外增益。这支持了第4.2节的结论。

回归统计：利用图2中逐步更新的回归统计，有助于分类和回归任务的有效多任务学习。通过对比表3中是否使用该统计的模型，可以观察到其带来的益处。学习过程对这些统计并不敏感。

阶段数量：阶段数量的影响总结在表4中。增加第二个检测阶段显著提升了基线检测器的性能。三个检测阶段仍能带来显著改进，但添加第四个阶段（ $u = 0.75$ ）会导致性能轻微下降。然而需要注意的是，虽然整体AP性能有所降低，但四阶段级联在高IoU水平下表现最佳。三阶段级联实现了最佳平衡。

5.5. 与现有技术的比较

基于FPN+和ResNet-101骨干网络的Cascade R-CNN，在表5中与最先进的*single-model*目标检测器进行了比较。设置如第5.1.1节所述，但总共进行了280k次训练迭代，学习率在160k和240k次迭代时下降。RoI的数量也增加到了512。表5中的第一组检测器是单阶段检测器，第二组是两阶段检测器，最后一组是多阶段检测器（Cascade R-CNN采用3阶段+RPN）。所有对比的最先进检测器均使用 $u = 0.5$ 进行训练。

	backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
YOLOv2 [26]	DarkNet-19	21.6	44.0	19.2	5.0	22.4	35.5
SSD513 [23]	ResNet-101	31.2	50.4	33.3	10.2	34.5	49.8
RetinaNet [22]	ResNet-101	39.1	59.1	42.3	21.8	42.7	50.2
Faster R-CNN+++ [16]*	ResNet-101	34.9	55.7	37.4	15.6	38.7	50.9
Faster R-CNN w FPN [21]	ResNet-101	36.2	59.1	39.0	18.2	39.0	48.2
Faster R-CNN w FPN+ (ours)	ResNet-101	38.8	61.1	41.9	21.3	41.8	49.8
Faster R-CNN by G-RMI [17]	Inception-ResNet-v2	34.7	55.5	36.7	13.5	38.1	52.0
Deformable R-FCN [5]*	Aligned-Inception-ResNet	37.5	58.0	40.8	19.4	40.1	52.5
Mask R-CNN [14]	ResNet-101	38.2	60.3	41.7	20.1	41.1	50.2
AttractionNet [10]*	VGG16+Wide ResNet	35.7	53.4	39.3	15.6	38.0	52.7
Cascade R-CNN	ResNet-101	42.8	62.1	46.3	23.7	45.5	55.2

Table 5. Comparison with the state-of-the-art *single-model* detectors on COCO *test-dev*. The entries denoted by “*” used bells and whistles at inference.

	backbone	cascade	train speed	test speed	param	val (5k)						test-dev (20k)					
						AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Faster R-CNN	VGG	✗	0.12s	0.075s	278M	23.6	43.9	23.0	8.0	26.2	35.5	23.5	43.9	22.6	8.1	25.1	34.7
		✓	0.14s	0.115s	704M	27.0	44.2	27.7	8.6	29.1	42.2	26.9	44.3	27.8	8.3	28.2	41.1
R-FCN	ResNet-50	✗	0.19s	0.07s	133M	27.0	48.7	26.9	9.8	30.9	40.3	27.1	49.0	26.9	10.4	29.7	39.2
		✓	0.24s	0.075s	184M	31.1	49.8	32.8	10.4	34.4	48.5	30.9	49.9	32.6	10.5	33.1	46.9
R-FCN	ResNet-101	✗	0.23s	0.075s	206M	30.3	52.2	30.8	12.0	34.7	44.3	30.5	52.9	31.2	12.0	33.9	43.8
		✓	0.29s	0.083s	256M	33.3	52.0	35.2	11.8	37.2	51.1	33.3	52.6	35.2	12.1	36.2	49.3
FPN+	ResNet-50	✗	0.30s	0.095s	165M	36.5	58.6	39.2	20.8	40.0	47.8	36.5	59.0	39.2	20.3	38.8	46.4
		✓	0.33s	0.115s	272M	40.3	59.4	43.7	22.9	43.7	54.1	40.6	59.9	44.0	22.6	42.7	52.1
FPN+	ResNet-101	✗	0.38s	0.115s	238M	38.5	60.6	41.7	22.1	41.9	51.1	38.8	61.1	41.9	21.3	41.8	49.8
		✓	0.41s	0.14s	345M	42.7	61.6	46.6	23.8	46.2	57.4	42.8	62.1	46.3	23.7	45.5	55.2

Table 6. Detailed comparison on multiple popular baseline object detectors. All speeds are reported per image on a single Titan Xp GPU.

noted that our FPN+ implementation is better than the original FPN [21], providing a very strong baseline. In addition, the extension from FPN+ to Cascade R-CNN improved performance by ~ 4 points. The Cascade R-CNN also outperformed all *single-model* detectors by a large margin, under all evaluation metrics. This includes the *single-model* entries of the COCO challenge winners in 2015 and 2016 (Faster R-CNN+++ [16], and G-RMI [17]), and the very recent Deformable R-FCN [5], RetinaNet [22] and Mask R-CNN [14]. The best multi-stage detector on COCO, AttractionNet [10], used *iterative BBox* for proposal generation. Although many enhancements were used in AttractionNet, the vanilla Cascade R-CNN still outperforms it by 7.1 points. Note that, unlike Mask R-CNN, no segmentation information is exploited in the Cascade R-CNN. Finally, the vanilla *single-model* Cascade R-CNN also surpasses the heavily engineered ensemble detectors that won the COCO challenge in 2015 and 2016 (AP 37.4 and 41.6, respectively)¹.

5.6. Generalization Capacity

Three-stage Cascade R-CNN of all three baseline detectors are compared in Table 6. All settings are as above, with the changes of Section 5.5 for FPN+.

¹<http://cocodataset.org/#detections-leaderboard>

Detection Performance: Again, our implementations are better than the original detectors [27, 4, 21]. Still, the Cascade R-CNN improves on these baselines consistently by 2 $\sim$ 4 points, independently of their strength. These gains are also consistent on *val* and *test-dev*. These results suggest that the Cascade R-CNN is widely applicable across detector architectures.

Parameter and Timing: The number of the Cascade R-CNN parameters increases with the number of cascade stages. The increase is linear in the parameter number of the baseline detector heads. In addition, because the computational cost of a detection head is usually small when compared to the RPN, the computational overhead of the Cascade R-CNN is small, at both training and testing.

6. Conclusion

In this paper, we proposed a multi-stage object detection framework, the Cascade R-CNN, for the design of high quality object detectors. This architecture was shown to avoid the problems of overfitting at training and quality mismatch at inference. The solid and consistent detection improvements of the Cascade R-CNN on the challenging COCO dataset suggest the modeling and understanding of various concurring factors are required to advance object

	backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
YOLOv2 [26]	DarkNet-19	21.6	44.0	19.2	5.0	22.4	35.5
SSD513 [23]	ResNet-101	31.2	50.4	33.3	10.2	34.5	49.8
RetinaNet [22]	ResNet-101	39.1	59.1	42.3	21.8	42.7	50.2
Faster R-CNN+++ [16]*	ResNet-101	34.9	55.7	37.4	15.6	38.7	50.9
Faster R-CNN w FPN [21]	ResNet-101	36.2	59.1	39.0	18.2	39.0	48.2
Faster R-CNN w FPN+ (ours)	ResNet-101	38.8	61.1	41.9	21.3	41.8	49.8
Faster R-CNN by G-RMI [17]	Inception-ResNet-v2	34.7	55.5	36.7	13.5	38.1	52.0
Deformable R-FCN [5]*	Aligned-Inception-ResNet	37.5	58.0	40.8	19.4	40.1	52.5
Mask R-CNN [14]	ResNet-101	38.2	60.3	41.7	20.1	41.1	50.2
AttractionNet [10]*	VGG16+Wide ResNet	35.7	53.4	39.3	15.6	38.0	52.7
Cascade R-CNN	ResNet-101	42.8	62.1	46.3	23.7	45.5	55.2

表5. 在COCO test-dev上与最先进的single-model检测器对比。标有“*”的条目在推理时使用了附加技巧。

		backbone	cascade	train speed	test speed	param	val (5k)						test-dev (20k)					
							AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Faster R-CNN	VGG	✗		0.12s	0.075s	278M	23.6	43.9	23.0	8.0	26.2	35.5	23.5	43.9	22.6	8.1	25.1	34.7
		✓		0.14s	0.115s	704M	27.0	44.2	27.7	8.6	29.1	42.2	26.9	44.3	27.8	8.3	28.2	41.1
R-FCN	ResNet-50	✗		0.19s	0.07s	133M	27.0	48.7	26.9	9.8	30.9	40.3	27.1	49.0	26.9	10.4	29.7	39.2
		✓		0.24s	0.075s	184M	31.1	49.8	32.8	10.4	34.4	48.5	30.9	49.9	32.6	10.5	33.1	46.9
R-FCN	ResNet-101	✗		0.23s	0.075s	206M	30.3	52.2	30.8	12.0	34.7	44.3	30.5	52.9	31.2	12.0	33.9	43.8
		✓		0.29s	0.083s	256M	33.3	52.0	35.2	11.8	37.2	51.1	33.3	52.6	35.2	12.1	36.2	49.3
FPN+	ResNet-50	✗		0.30s	0.095s	165M	36.5	58.6	39.2	20.8	40.0	47.8	36.5	59.0	39.2	20.3	38.8	46.4
		✓		0.33s	0.115s	272M	40.3	59.4	43.7	22.9	43.7	54.1	40.6	59.9	44.0	22.6	42.7	52.1
FPN+	ResNet-101	✗		0.38s	0.115s	238M	38.5	60.6	41.7	22.1	41.9	51.1	38.8	61.1	41.9	21.3	41.8	49.8
		✓		0.41s	0.14s	345M	42.7	61.6	46.6	23.8	46.2	57.4	42.8	62.1	46.3	23.7	45.5	55.2

表6. 多个流行基线目标检测器的详细比较。所有速度均在单个Titan Xp GPU上按每张图像报告。

指出我们的FPN+实现优于原始FPN[21]，提供了非常强大的基线。此外，从FPN+扩展到Cascade R-CNN将性能提升了~4个百分点。在所有评估指标下，Cascade R-CNN也以显著优势超越了所有single-model检测器。这包括2015年和2016年COCO挑战赛获胜者的single-model参赛方案（Faster R-CNN+++[16]和G-RMI[17]），以及近期提出的Deformable R-FCN[5]、RetinaNet[22]和Mask R-CNN[14]。COCO上最佳的多阶段检测器AttractionNet[10]使用了iterative BBox进行候选框生成。尽管AttractionNet采用了多种增强技术，基础版Cascade R-CNN仍以7.1个百分点的优势超越它。需注意的是，与Mask R-CNN不同，Cascade R-CNN未利用任何分割信息。最后，基础版single-modelCascade R-CNN也超越了2015年和2016年COCO挑战赛中经过深度工程优化的集成检测器（AP分别为37.4和41.6）¹。

5.6. 泛化能力

所有三个基线检测器的三阶段级联R-CNN均在表6中进行比较。所有设置如上所述，仅针对FPN+采用了第5.5节的修改。

¹<http://cocodataset.org/#detections-leaderboard>

检测性能：同样，我们的实现优于原始检测器[27, 4, 21]。尽管如此，Cascade R-CNN在不同强度的基准上始终提升了2~4个点。这些增益在验证集和测试开发集上也是一致的。这些结果表明，Cascade R-CNN在检测器架构中具有广泛的适用性。

参数与时机：级联R-CNN的参数数量随级联阶段数增加而增加，其增长量与基准检测头参数数量呈线性关系。此外，由于检测头的计算成本通常远小于区域提议网络（RPN），因此级联R-CNN在训练和测试阶段的计算开销都很小。

6. 结论

本文提出了一种多阶段目标检测框架——Cascade R-CNN，用于设计高质量目标检测器。该架构被证明能够避免训练过程中的过拟合问题以及推理时的质量不匹配问题。Cascade R-CNN在具有挑战性的COCO数据集上取得坚实而一致的检测效果提升，表明要推动目标检测发展，需要对各种并发因素进行建模与理解。

detection. The Cascade R-CNN was shown to be applicable to many object detection architectures. We believe that it can be useful to many future object detection research efforts.

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检测。Cascade R-CNN被证明适用于许多目标检测架构。我们相信它能为未来许多目标检测研究工作提供帮助。

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